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Promotion of Nb_2O_5 on the wustite-based iron catalyst for ammonia synthesis

Wenfeng Han*, Shiliang Huang, Tianhong Cheng, Haodong Tang, Ying Li, Huazhang Liu*

Institute of Industrial Catalysis, Zhejiang University of Technology, Hangzhou, Zhejiang 310032, PR China

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ABSTRACT

Niobium was selected and investigated as a potential promoter for wustite-based catalyst (WBC) for ammonia synthesis. Experiments on reduction performance, activity test and H_2-TGA, in situ XRD as well as XPS were carried out to obtain the promotion effect and mechanism involved. Niobium as a promoter was confirmed to enhance the reduction of WBC significantly. This behavior is highly desired for industry in terms of catalyst regeneration and lesser pretreatment time for fabrication regardless the unimproved catalytic performance for Nb_2O_5-doped wustite-based catalyst (Nb-WBC). Possible reasons for these phenomena are discussed. It is suggested that Nb_2O_5 is not favorable for the segregation or formation of solid solutions on the catalyst surface, which are difficult to be reduced. However, it seems that niobium does not promote the growth of [2 1 1] plane, which is active for ammonia synthesis.

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1. Introduction

Since Haber and Bosch developed a practical process of ammonia synthesis, catalysts for ammonia synthesis have been of industrial and theoretical importance [1]. Mainly, the development of fused iron catalysts experiences two stages. Initially iron catalysts are based on magnetite, and an interesting model structure describing these magnetite-based catalysts were reported by Fagherazzi et al. [2]. Later in 1986, Liu et al. discovered a new type of catalysts based on wustite (WBC). This is an impressive example, the first significant innovation in the industrial iron-based ammonia synthesis catalyst after the development of catalysts for the Haber–Bosch process in 1910. The WBC shows much higher activity and lower reduction temperature than the traditional magnetite-based catalysts (MBC) [3,4]. Upon this discovery, a reconsideration of the present consolidated knowledge on iron ammonia synthesis catalyst was suggested [5,6]. As the same with MBC, the effect of promoter on performance of wustite-based catalyst is very significant, and there is a great potential for further improvement of WBC [7]. Although almost all the elements in the Periodical table were investigated, there is, however, the promoting effect of niobium on iron catalyst rarely reported.

The function of niobium compounds in catalysis can be either a promoter or an active phase, support, solid acid catalyst, or redox material [8]. Because a supported metal oxide often shows different catalytic properties from those of the bulk oxide, much research on supported niobium oxide has been carried out in attempting to control the catalytic properties and the structures of the surface niobium species over different supporting metal oxides [9–11]. Although not for ammonia synthesis, it was also reported that niobium increases the catalytic activity and stability of iron oxides catalysts [12–14]. Indeed, niobium brings interesting features to this system such as ionic compatibility with the iron oxide structure and higher reactivity [12]. Oliveira et al. reported that hematites, where Fe^{3+} ions are partially replaced by Nb^{5+}, are active and selective catalysts for the dehydration of isopropanol to propene [15]. It was also demonstrated that it is possible to activate hematite with niobium and a further catalytic improvement can be obtained with the formation of peroxo groups, which presents a high oxidizing power. It prompts the investigation of niobium on iron-based catalysts for ammonia synthesis.

In the present study, niobium oxide was adopted as a promoter of fused iron catalysts and its effect on reduction behavior and the activities for ammonia synthesis were investigated. The catalyst promoted with niobium oxide showed improved performance of reduction than that of WBC. These catalysts were further characterized by in situ X-ray diffraction, thermogravimetric measurements, and X-ray photoelectron spectroscopy. Based on these results, the influence of niobium oxide on the reduction behavior and the activity of fused iron catalyst were discussed.

* Corresponding author. Tel.: +86 571 88320063; fax: +86 571 88320259.
E-mail addresses: hanwf@zjut.edu.cn (W. Han), cuihua@zjut.edu.cn (H. Liu).

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Table 1
Chemical composition of natural magnetite, WBC and Nb-WBC catalysts.

Samples	Chemical composition (wt%)							Fe^{2+}/Fe^{3+}
	FeO	Fe_2O_3	Al_2O_3	K_2O	CaO	Nb_2O_5	Others	
Natural magnetite	26.25	72.92	0.21	0.03	0.09	–	0.50	0.40
Nb-WBC	80.00	12.52	1.8	0.6	1.8	0.6	2.08	7.10
WBC	80.46	12.66	1.8	0.6	1.8	–	2.68	7.06

2. Experimental

2.1. Catalyst preparation

The wustite catalysts were prepared by mixing appropriate proportions of purified magnetite, iron powder, alumina, potassium nitrate, calcium carbonate and other promoters, and melting in an electric resistance furnace at about 1600 °C. Melting processes were carried out over the equipment composed with a stainless steel container, a water-cooled electrodes and a cooler with water jacket. The Fe^{2+}/Fe^{3+} ion ratio in wustite-based catalysts was controlled by addition amount of iron powder as reducing agent. Following melting of magnetite, iron powder and promoters, the melt was poured out and allowed to be water-cooled. For Nb_2O_5 doped WBC, 0.6 wt% of Nb_2O_5 was added together with other promoters. Then, by crushing and screening out, the irregular catalyst was obtained, and particles with diameters in the range from 1.0 to 1.4 mm were collected. The chemical compositions of natural magnetite, WBC and Nb-WBC adopted in this study are listed in Table 1.

2.2. Activity test for ammonia synthesis

Ammonia synthesis reaction over the samples were investigated in a fixed-bed reactor with an inside diameter of 14 mm in the temperature range of 400–500 °C, space velocity of 30,000 h^{-1} and pressure of 15 MPa. The catalyst bed was loaded in the isothermal zone of the reactor, and both the ends of the catalyst bed were filled with quartz with the size range of 1.0–2.0 mm to prevent the ammonia synthesis gas from channeling. The catalysts were reduced by the mixture of N_2 and H_2 (H_2:N_2 = 3:1) at a pressure of 5.0 MPa and with a space velocity of 30,000 h^{-1} at 400, 425, 450, 475 and 500 °C for 2, 6, 10, 4 and 2 h, respectively. The synthesis gas was derived from the decomposition of ammonia with deep removal of H_2O, CO, CO_2 and residual NH_3 over Pd, 13X, and 5A molecular sieves and compressed by a compressor.

2.3. Catalysts characterization

Thermogravimetric analysis in hydrogen (H_2-TGA) was conducted on NETZSCH STA449C. About 10 mg of catalyst (100–300 standard mesh) were adopted, and hydrogen (30 ml/min) is purified with Pd, silicone and 5A molecular sieves. Prior to the experiments, temperature was increased to 200 °C at a ramp rate of 10 °C/min and then was hold at 200 °C for 1 h to remove physically adsorbed water in Ar atmosphere. After cooled down to room temperature, it was increased to 650 °C at a heating rate of 3 °C/min.

X-ray diffraction (XRD) was performed with a Thermo ARL X'TRA diffractometer using Cu-Kα radiation (λ = 0.154056 nm), equipped with a Si(Li) solid detector at 40 kV/40 mA with a monochromator. The in situ experiments were carried out in Anton Paar XRK 900 reactor built in the diffractometer with a TCU 750 temperature control unit, the temperature error is within ±1 °C. The in situ XRD performed at a stipulated heating rate in the range of 25–500 °C with 60 ml/min of H_2 and 101 kPa of pressure. The XRD patterns tracing the phase transformation are carried out in a continuous scan mode, with a step of 0.02° from 30° to 90° at a speed of 2.5°/min. The crystallite size for the active phase α-Fe in both (2 1 1), (2 0 0) and (1 1 0) directions was calculated using the Scherrer equation ($d = K\lambda/\beta\cos\theta$, where, K-Scherrer equation constant = 0.89, λ-wavelength of radiation, β-line broadening of the XRD peak, and θ-diffraction angle.).

Temperature programmed desorption (H_2-TPD) was performed on an in situ reaction device equipped with a MS detector. 100 mg sample was pretreated in H_2 stream at 450 °C for 5 h. Then the sample was cooled down to room temperature in H_2 atmosphere. Prior to desorption experiment, the sample was purged with He for 1 h to remove physically adsorbed species. Then, the furnace temperature was gradually increased to 800 °C with a ramp rate of 5 °C/min. The desorption gases were recorded by a mass spectrometer.

For X-ray photoelectron spectroscopy (XPS) experiments, a spectrometer from Kratos AXIS Ultra DLD photoelectron spectroscopy with a monochromatized microfocused Al X-ray source was employed. Charging of samples was corrected by setting the binding energy of adventitious carbon (C 1s) at 284.6 eV. Prior to the measurements, the powder sample, pressed into self-supporting disks, was loaded into a sub-chamber and then evacuated at 25 °C for 4 h.

3. Results and discussion

3.1. Reduction behavior of catalysts

As reported by Pernicone, reduction of wustite catalysts is much faster than that of magnetite [5]. According to the temperature programmed reduction, compared with magnetite catalyst, the reduction peak of wustite catalyst shifts to lower temperature [16]. Therefore, during their industrial applications, wustite catalyst has better reducibility and shorter pretreatment time, which is favorable for the ammonia plants.

The reduction performance of wustite catalyst can be further promoted by the dopping of Nb_2O_5. Thermogravimetric technique under hydrogen atmosphere (H_2-TGA) was adopted to evaluate the reduction rate of wustite iron catalysts. The H_2-TGA and DTG profiles of WBC and Nb-WBC are presented in Fig. 1. The weight loss of WBC and Nb-WBC are 20.8% and 20.1%, respectively. According to the Fe^{2+}/Fe^{3+} of 7.06 (or 7.10 Nb-WBC) and promoter contents as illustrated in Table 1, the calculated weight loss of WBC should be around 21.68% (and 21.53% for Nb-WBC). During catalysts preparation, small amounts of FeO or Fe_2O_3 underwent reaction with Al_2O_3 or CaO et al. forming solid solutions, which are difficult to be reduced. Therefore, the experimental value is very close to the weight loss of calculation indicating the nearly complete reduction of catalyst precursors.

According to the results of DTG in Fig. 1, reduction peaks of WBC and Nb-WBC center around 475 °C and 450 °C respectively. Clearly, the doping of niobium promotes the reduction of WBC by reduction temperature of 25 °C lower. In addition, at these temperatures, the corresponding reduction rates are 0.76%/min and 0.88%/min for WBC and Nb-WBC, respectively. The reduction rate is increased by 16% following Nb doping. For WBC, the starting reduction temperature is about 387.5 °C and the end temperature is about 505.5 °C, while the corresponding temperatures for Nb-WBC are 363.5 °C and 480.5 °C, respectively. They are 24 °C and 25 °C lower than those of WBC, which agrees well with the peak temperature difference.

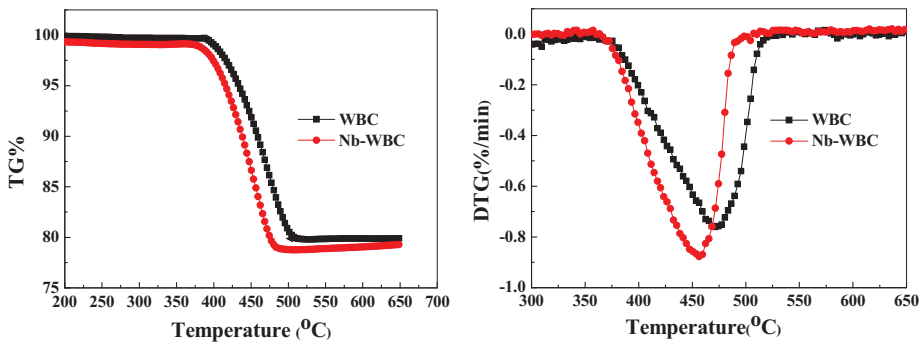

Fig. 1. H_2-TGA (left) and DTG (right) profiles of WBC and Nb-WBC catalysts.

Table 2
Ammonia concentration of WBC and Nb-WBC as a function of time during reduction.

Temperature/°C	Times on line/h	Ammonia concentration/%NH_3	
		WBC	Nb-WBC
425	2	2.84	4.14
	3	4.42	5.69
	4	5.85	7.31
	5	6.76	7.59
	6	7.20	8.03
	7	7.38	8.07
450	9	7.54	8.04
	12	7.65	8.06
	15	8.06	8.17
	17	7.83	8.30
475	20	6.78	6.92
	22	6.82	6.86

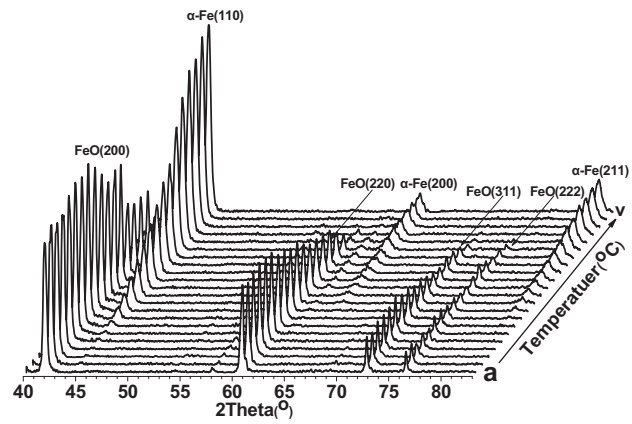

Fig. 2. The evolution of diffraction patterns of WBC sample as a function of reduction temperature. Temperature (°C): (a) RT, (b) 100, (c) 200, (d) 250, (e) 300, (f) 325, (g) 330, (h) 335, (i) 340, (j) 345, (k) 350, (l) 355, (m) 360, (n) 365, (o) 370, (p) 375, (q) 390, (r) 400, (s) 410 (t) 415, (u) 425, (v) 450.

The results confirm that the niobium-doped catalyst has a lower reduction temperature and a faster reduction rate than WBC.

During reduction, once the α-Fe forms, ammonia synthesis reaction commences simultaneously. Therefore, ammonia concentration during reduction may function as an indicator of the reduction reaction. In addition, as we all know that ammonia synthesis is an exothermal reaction, which is favorable for the endothermal reduction reaction.

The ammonia concentrations during reduction of WBC and Nb-WBC catalysts were compared at temperatures between 400 and 475 °C in the flow of N_2 and H_2 mixture (H_2:N_2 = 3:1) at a pressure of 5.0 MPa and with a space velocity of 30,000 h^{-1}. The ammonia concentrations of WBC and Nb-WBC during reduction as a function of time are shown in Table 2.

As can be seen from Table 2, following reduction at 425 °C for 6 h, ammonia concentration of Nb-WBC is as high as 8% and keeps almost unchanged at temperatures below 475 °C indicating the complete reduction during this period. Due to the limit of equilibrium, ammonia concentration drops at 475 °C. By contrast, at this reduction time, the ammonia concentration of WBC is only 7.2%. It needs further reduction of another 9 h at 450 °C to achieve ammonia concentration of 8%. Clearly, reduction time of Nb-WBC is at least 9 h shorter than that of WBC (60% shorter). This behavior is highly desired for industry in terms of catalyst regeneration and lesser pretreatment time for fabrication regardless the unimproved catalytic performance for Nb-WBC catalyst.

As illustrated in Table 2, at the initial stage of reduction, ammonia concentration is nearly doubled. Clearly, small amounts of Nb_2O_5 play a significant role in the reduction of wustite catalyst. Higher ammonia concentration during reduction suggests faster formation of α-Fe, which is the active phase of ammonia synthesis. Under the conditions studied, the activity during reduction follows the similar trend with reduction rate (H_2-TGA results). We suggest that growth and restructuring of formed α-Fe play a minor role in activity. In addition, although Nb-WBC catalyst shows faster reduction rate than that of WBC catalyst, they have the similar reduction degree (precursors of both catalysts were completely reduced finally) and hence they exhibit similar activity at the final stage of reduction.

3.2. *In situ X-ray diffraction*

The in situ X-ray diffraction patterns of WBC and Nb-WBC catalysts as a function of reduction temperature are shown in Figs. 2 and 3, respectively. During reduction, only the active phase α-Fe for ammonia synthesis was observed without other intermediates. Apparently, for both two catalysts, α-Fe forms directly from the reduction of, wustite ($Fe_{1-x}O$) precursor, which is consistent with our previous study [17,18]. As illustrated in Fig. 2, the [1 0 0] peak of the active phase α-Fe is detectable at about 325 °C indicating reduction of WBC catalyst commences at this temperature. Following reduction, the peak intensity of precursor drops with reduction temperature, and it disappears at about 415 °C completely. Meanwhile, three major Fe faces, namely [1 1 0], [2 0 0] and [2 1 1] are recorded. Unfortunately, Fe [1 1 1] is not detectable because of the limitation of instrument.

Following the addition of small amounts of Nb_2O_5, in situ X-ray diffraction patterns (see Fig. 3) exhibit similar trend with that of WBC. However, compared with the details between these two catalysts, it can be found that the reduction temperatures of niobium-doped catalyst are at the range of 310 °C and 395 °C, which are 15 °C and 20 °C lower than those of WBC, respectively.

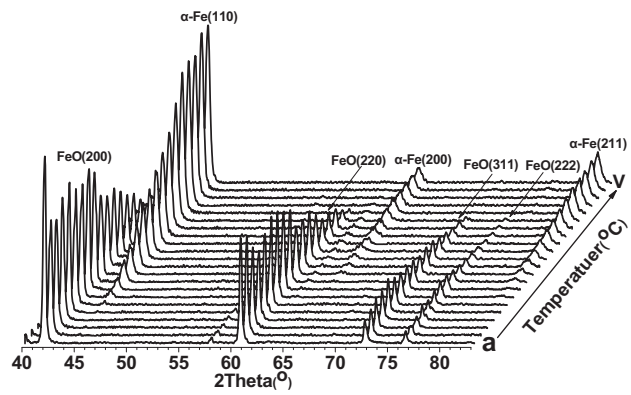

Fig. 3. The evolution of diffraction patterns of Nb-doped sample as a fuction of reduction temperature. Temperature (°C): (a) RT, (b) 100, (c) 200, (d) 250, (e) 300, (f) 310, (g) 320, (h) 325, (i) 330, (j) 335, (k) 340, (l) 345, (m) 350, (n) 355, (o) 360, (p) 365, (q) 375, (r) 390, (s) 395, (t) 400, (u) 425, (v) 450.

During reduction, the growth of α-Fe crystallites is anisotropy with preferential growth along the [1 1 0], [2 0 0] and [2 1 1] crystallographic directions with the intensities of these reflection peaks increasing with temperature. To further investigate the growth behavior of different planes, the evolution of planes [2 1 1], [2 0 0] and [1 1 0] reflection is compared. The ratio of reflection intensities for [2 1 1]/[1 1 0] and [2 0 0]/[1 1 0] are presented in Fig. 4. At low temperatures, the reduction is not very evident and therefore the values of ratio are close to 1. However, at higher temperatures, $D_{[2\,1\,1]/[1\,1\,0]}$ and $D_{[2\,0\,0]/[1\,1\,0]}$ decrease with temperature significantly. According to XRD, $D_{[2\,1\,1]/[1\,1\,0]}$ and $D_{[2\,0\,0]/[1\,1\,0]}$ represent the growth extents, and it relates the active site. The lower value implies the higher concentration of [2 1 1] and [2 0 0] planes.

As Fig. 4 shows, the growth of [2 1 1] and [2 0 0] planes increases with temperature. The doping of Nb_2O_5 enables the shift of the initial points toward low temperatures by 15–20 °C, which is consistent with our previous observation. In addition, $D_{[2\,1\,1]/[1\,1\,0]}$ and $D_{[2\,0\,0]/[1\,1\,0]}$ of Nb_2O_5 doped catalyst are smaller than that of WBC catalyst during major reduction stage (310–400 °C), indicating that [2 1 1] and [2 0 0] planes of Nb_2O_5 doped catalyst are better developed. For Fe_3O_4-based catalyst, the value of D[2 1 1]/[1 1 0] was reported to be 0.86 [17], while for WBC ($Fe_{1-x}O$-based catalyst) is around 0.72 after reduction. Hence, WBC possesses more developed [2 1 1] and [2 0 0] planes. Following the doping of Nb_2O_5, although the growth rates of [2 1 1] and [2 0 0] planes are enhanced, their amounts are relatively lower than that of un-doped (value of $D_{[2\,1\,1]/[1\,1\,0]}$ is about 0.74 for Nb_2O_5 doped catalyst).

3.3. H_2-TPD-MS (H_2 programmed desorption)

H_2-TPD is served as a key analysis method for the relative number of active sites in iron catalysts for ammonia synthesis [19]. We have carried out systematic investigation on H_2-TPD performed for WBC catalyst [20]. The precursor, namely Fe_3O_4, $Fe_{1-x}O$ or the mixture of Fe_3O_4 and $Fe_{1-x}O$ affected the H_2 desorption during experiments. For traditional Fe_3O_4 based catalyst, there are three H_2 desorption peaks (α_1, α_2 and α_3) observed with temperatures at 130 °C, 300 °C and 450 °C, respectively [19]. However, for $Fe_{1-x}O$ based (WBC) catalyst, as shown in Fig. 6, only two desorption peaks are detected whose temperatures are 436 °C (α_2) and 555 °C(α_3) [20]. Clearly, α_2 is more likely to be connected with ammonia synthesis since their temperatures are close from each other.

It has been long noticed that ammonia synthesis is significantly inhibited by the strong adsorption of hydrogen over Ru catalyst [21–24]. Similarly, strong adsorption hydrogen species over Fe catalysts has unfavorable effect on catalytic performance. Therefore, we suggest that hydrogen species (α_3) plays a negative role in ammonia synthesis as it strongly occupies the active sites restraining the dissociate adsorption of N_2, which is the rate-determining step for ammonia synthesis.

As Fig. 5 shows, the small peak with desorption temperature of 555 °C indicates the small amounts of α_3 species over WBC catalyst. Therefore, the inhibition effect of strong hydrogen adsorption

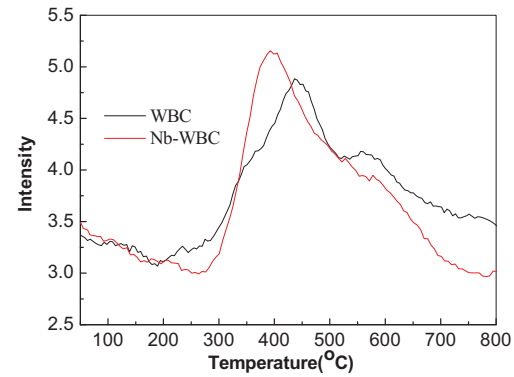

Fig. 5. H_2-TPD-MS profiles of WBC and Nb-WBC catalysts.

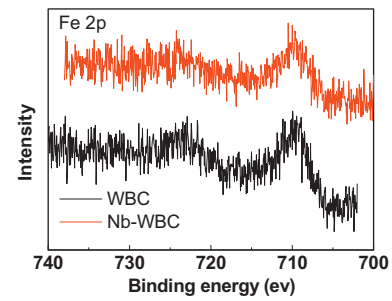

Fig. 6. The XPS Fe 2p core level spectra of WBC and Nb-WBC thin films.

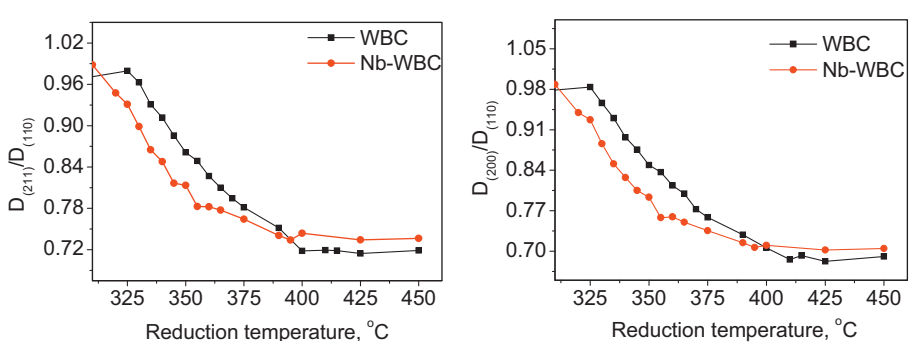

Fig. 4. The ratio of reflection intensities for [2 1 1]/[1 1 0] and [2 0 0]/[1 1 0] during in situ XRD.

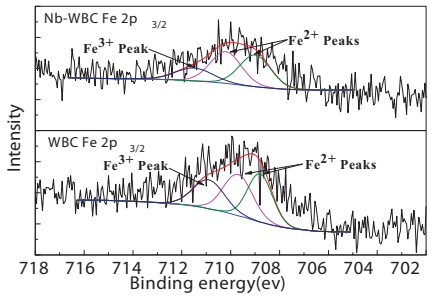

Fig. 7. Deconvoluted XPS Fe $2p_{3/2}$ profiles. (a) WBC thin film and (b) Nb-WBC thin film.

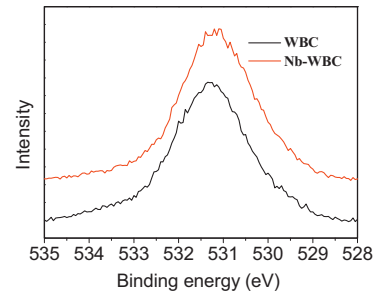

Fig. 8. The XPS O 1s core level spectra of WBC and Nb-WBC thin films.

over WBC is negligible. In addition, following the doping of Nb_2O_5, the peak intensity further declines implying that relatively smaller α_3 exists over Nb-WBC. Meanwhile, Following Nb_2O_5 doping, the peak intensity of α_2 increases noticeably indicating the enhancement of active sites for N_2 dissociation. Also it is noted that the desorption temperature is 40 °C lower than that of WBC. During reaction, as mention above, these species are capable of interacting with N_2 forming ammonia [20]. Hence, with less inhibition effect of strong hydrogen adsorption and lower desorption temperature of adsorbed H_2 species, higher activity than WBC catalyst is expected following Nb_2O_5 doping. However, the doping of Nb_2O_5 facilitates the sintering of active phase, which contributes to the low activity (see discussion of Section 3.4).

3.4. X-ray photoelection spectroscopy (XPS)

To further investigate the effect of Nb_2O_5 on the WBC wustite catalyst for ammonia synthesis, XPS experiments were carried out. Fig. 6 presents the XPS data on Fe 2p core levels of WBC and Nb_2O_5 doped catalysts. Due to the spin–orbit coupling, the Fe 2p core levels split into $2p_{3/2}$ and $2p_{1/2}$ components. Hence, two peaks with binding energy (BE) located at 711 eV and 724 eV are related to the binding energies of Fe $2p_{3/2}$ and Fe $2p_{1/2}$ for iron oxides respectively [25,26].

Fig. 7a displays the details of Fe $2p_{3/2}$ profiles for WBC catalyst. Clearly, Fe $2p_{3/2}$ high resolution spectra can be well deconvoluted into three peaks, situated at binding energies of 711 eV, 710 eV and 709 eV respectively which is consistent with Grosvenor's results [27]. Therefore, both Fe^{3+} and Fe^{2+} exist in the catalyst although WBC is desired to obtain FeO precursor (Wustite). In addition, splitting Fe^{2+} peaks at 710 eV and 709 eV indicate that there are two Fe^{2+} environments for the catalyst precursors. WBC is a catalyst with multi-promoters accounting for about 7% of total catalyst weight. Either the structural promoters or the electric promoters interact with Fe differing it from the bulk Fe in XPS spectrum.

According to the spectrum, significant amounts of Fe^{3+} were observed for wustite catalyst. During the preparation of WBC catalyst, the mixture of Fe_3O_4, Fe as well as Al, K, Ca and Mg promoters in the form of Al_2O_3, KNO_3, $CaCO_3$ and $MgCO_3$ were melted in an electric furnace (around 1500–1600 °C). Different from the preparation of traditional Fe_3O_4 based catalyst, the catalyst precursors are obtained via the reactions between Fe and Fe_3O_4/Fe_2O_3 (see reactions R1 and R2). Although their ratios are determined by these reactions, small amounts of Fe^{3+} residue are acceptable.

$$4Fe_2O_3 + Fe \rightarrow 3Fe_3O_4 \quad (R1)$$

$$Fe_3O_4 + Fe \rightarrow 4FeO \quad (R2)$$

In addition, various solid solutions were expected under these extreme conditions via reactions as follow [28]. They are also responsible for the presence of Fe^{3+} species in the catalyst precursor.

$$FeO + Al_2O_3 \rightarrow FeAl_2O_4 \quad (R3)$$

$$Fe_2O_3 + K_2O \rightarrow 2KFeO_2 \quad (R4)$$

$$Fe_2O_3 + CaO \rightarrow CaFe_2O_4 \quad (R5)$$

$$Fe_2O_3 + MgO \rightarrow MgFe_2O_4 \quad (R6)$$

Following the doping of Nb_2O_5, as Fig. 7b exhibits, similar Fe^{2+} and Fe^{3+} states can be derived. However, the intensity of Fe^{3+} peak is relatively lower than that of WBC catalyst. According to the intensity, Fe^{2+}/Fe^{3+} is found to be 4.1 for Nb_2O_5 doped catalyst, while 2.7 is derived for WBC catalyst. As listed in Table 1, the Fe^{2+}/Fe^{3+} of bulk catalyst is around 7.1 for both catalysts. Clearly more Fe^{3+} presents over catalyst surface. As discussed above, significant amounts of solid solution may form during catalyst preparation. The segregation results in the enrichment of promoters as well as Fe^{3+} over the catalyst surface. It can be concluded that Nb_2O_5 partially inhibits the formation or segregation of these solid solution. As we all know, solid solutions are more difficult to be reduced than FeO during the activation or reaction operation of the catalysts [29]. Hence, improved reduction is expected which is well agreed with the results of TG and in situ XRD experiments.

The O 1s core level spectra of WBC and Nb_2O_5 doped catalysts are exhibited in Fig. 8. Usually, the binding energy of O in Fe_2O_3, Fe_3O_4 and FeO situates at around 530 eV [27]. Due to the segregation of promoters, significant amounts of O bind to Al, K and Ca. Consequently, shift of binding energy toward high energy is observed over WBC catalyst (531.35 eV). As argued above, presence of Nb_2O_5 hinders the formation or segregation of solid solutions. As a result, binding energy of O 1s over Nb-WBC (531.10 eV) is lower than that of WBC. It further confirms the influence of Nb_2O_5 on the segregation or formation of solid solutions.

Core level spectra of other elements, such as Ca 2p, K 2p, Al 2p and Mg 2p together with Fe 2p and O 1s are summarized in Table 3. Following the doping of Nb_2O_5, in addition to the BE shifts of Fe 2p and O 1s, BEs of Al 2p and Mg 2p also shift toward high energy. This is consistent with the conclusion that Nb_2O_5 is not favorable

Table 3
Binding energy of Ca 2p, K 2p, Al 2p and Mg 2p and their concentration on the surface of catalyst.

	WBC		Nb-WBC	
	BE(eV)	Mass content (%)	BE (eV)	Mass content (%)
O 1s	531.35	53.44	531.10	51.56
Fe 2p	710.80	3.60	711.15	4.51
Ca 2p	346.95	7.05	346.90	7.07
K 2p	292.70	16.58	292.65	15.60
Al 2p	73.60	13.17	73.85	15.89
Mg 2p	55.30	5.20	55.50	5.02
Nb 3d	–	–	206.85	0.34
Others		0.96		0.01

Table 4
The diameter of predominant crystalline planes by Seherrer formula at different temperatures.

Temperature (°C)	Grain size of WBC (nm)			Temperature (°C)	Grain size of Nb-WBC (nm)		
	110	200	211		110	200	211
330	18.2	–	–	320	18.0	–	–
350	17.6	11.1	16.8	350	17.9	16.1	16.3
360	16.5	11.2	15.2	360	17.8	11.2	15.8
375	17.2	15.4	14.0	375	18.5	12.8	15.0
390	16.9	12.0	14.0	390	18.2	12.3	15.8
400	17.2	13.0	15.4	400	18.5	13.5	15.0
425	17.9	14.1	15.5	425	19.6	15.9	17.2
450	19.2	14.1	17.8	450	20.5	15.0	16.6
475	18.8	13.6	17.2	475	20.3	15.9	19.7
500	18.5	14.1	17.0	500	20.7	15.4	19.7

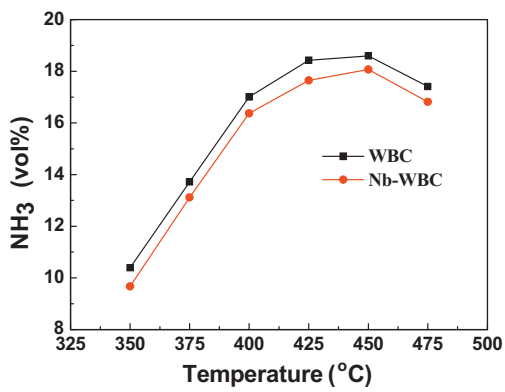

Fig. 9. Catalystic activities of WBC and Nb-WBC as a function of temperature.

for the segregation or formation of solid solutions on the catalyst surface.

As listed in Table 1, only 1.8 wt%, 0.6 wt% and 1.8 wt% of Al_2O_3, K_2O and CaO were introduced during catalysts preparation. However, according to the XPS data, the concentrations of Al_2O_3, K_2O and CaO are as high as 13.7%, 16.6% and 7.1%, respectively (see Table 3). It agrees with the segregation of promoters to the surface. By contrast, the loading amount of Nb_2O_5 is 0.6%, while only 0.34% was detected over the catalyst surface. Unlike other promoters, Nb_2O_5 does not tend to segregate to the surface. Due to the limitation of quantification accuracy for XPS, no significant differences in surface concentration of promoters between WBC and Nb-WBC are observed.

3.5. Catalystic activity of ammonia synthesis

Ammonia synthesis activities of the two samples were evaluated over a high pressure fixed bed reactor and the results are shown in Fig. 9. Among the reaction temperatures studied, the activity of WBC is marginally higher than that of niobium-doped catalyst.

For fused iron catalyst for ammonia synthesis, the results of Bernhard [29] is an indication for a significant anisotropy of the activated iron particles. The non-simultaneous evolution of the iron diffracted intensity in different orientations with temperature is an obvious sign of the non-isotropic nature of the activated catalyst. Different planes of active α-Fe have their own growth performance. Furthermore, the grain size is an indicator of the growth along with each plane. The larger the grain size is, the thicker along with the perpendicular direction is [30], and hence less planes are formed. The [2 1 1], [2 0 0] and [1 0 0] iron surfaces of grain sizes for WBC and niobium-doped catalysts during reduction process are estimated by Seherrer Formula at different temperatures. As demonstrated in Table 4, the non-simultaneous evolutions of three iron reflections differ from intensity in different orientations. According to Somorjai [31], as a structure sensitive reaction, ammonia synthesis activity of iron planes follows the order of [1 1 1] > [2 1 1] > [1 0 0] > [2 1 0] > [1 1 0]. Unfortunately, Fe [1 1 1] is not accessible by XRD. In the present study, the particles size of Fe [2 1 1] and [1 0 0] surfaces calculated from the Sherrer Formula is compared between two catalysts. As indicated in Table 4, the active [2 1 1] and [1 1 0] plane sizes of WBC catalyst are relatively smaller than that of niobium-doped catalyst. As demonstrated previously, although the growth rates of [2 1 1] and [2 0 0] planes are enhanced after niobium doping, their amounts are relatively lower than that of un-doped (value of $D_{[2\,1\,1]/[1\,1\,0]}$ is about 0.74 for Nb_2O_5 doped catalyst while it is 0.72 for WBC). Apparently these two reflections of WBC are better developed than the catalyst doped with niobium. Hence, it leads to higher activity of ammonia synthesis for WBC as it has more active sizes.

In addition, as listed in Table 4, grain size of [2 1 1] for Nb_2O_5 doped catalyst sinters at 425 °C while 450 °C for un-doped catalyst. Clearly, Nb_2O_5 doped catalyst tends to sinter at high temperatures, which is consistent with the results of XPS results. XPS data show that Nb_2O_5 is not favorable for the segregation or formation of solid solutions on the catalyst surface, which are difficult to be reduced during reduction. In the absence of these solid solutions (probably they function as structural promoters), sintering of active phase is facilitated which also contributes the low activity. However, additional experiments show that low reduction temperature or reaction temperature did not lead to the enhanced activity for Nb_2O_5 doped catalyst even with the reduction temperature lower than 425 °C. As indicated in Fig. 9, Nb_2O_5 doped catalyst achieves high activity at temperatures around 425–450 °C which poses challenge in avoiding the sintering of catalyst. Further investigation needs to be carried out on these issues.

4. Conclusions

The promotion effect of niobium on wustite-based catalyst (WBC) for ammonia synthesis was investigated by TG, reduction, in situ XRD and XPS as well as activity test. It reveals that niobium enhances the reduction of WBC catalyst significantly. The results of TG experiments confirm that the niobium-doped catalyst has a lower reduction temperature (25 °C lower than that of WBC) and a faster reduction rate than the sample of WBC. According to in situ XRD, WBC possesses more developed [2 1 1] and [2 0 0] planes. Following the doping of Nb_2O_5, although the growth rates of [2 1 1] and [2 0 0] planes are enhanced, their amounts are relatively lower than that of un-doped catalyst. XPS data show that Nb_2O_5 is not favorable for the segregation or formation of solid solutions on the catalyst surface, which are difficult to be reduced during reaction. Hence, it finally leads to the facile reduction performance of niobium doped catalyst. Although Nb_2O_5 improves the reduction of WBC catalyst, the activity keeps almost unchanged (or even marginally lower). As a structure sensitive reaction, [1 1 1]

and [2 1 1] planes are favorable for ammonia synthesis. However, it seems that niobium does not promote the growth of these planes during reduction.

From the industrial point view, although niobium does not improve the activity, it has significant effect on the reduction performance. Therefore, during the industrial application, wustite catalyst doped with niobium has better reducibility and shorter pretreatment time, which is favorable for the ammonia plants.

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